

Weeks 2-10

K-1

Spoken instructions; matching, building, and small counts.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-4
03 Repeating secret machines	5-7
04 Around the ring	8-10
05 Tower cities	11-13
06 Code-breaking with guarantees	14-16
07 Take-away games	17-19
08 Rooks and piles	20-22
09 Bouncing paths	23-25
10 Bridges and delivery routes	26-28

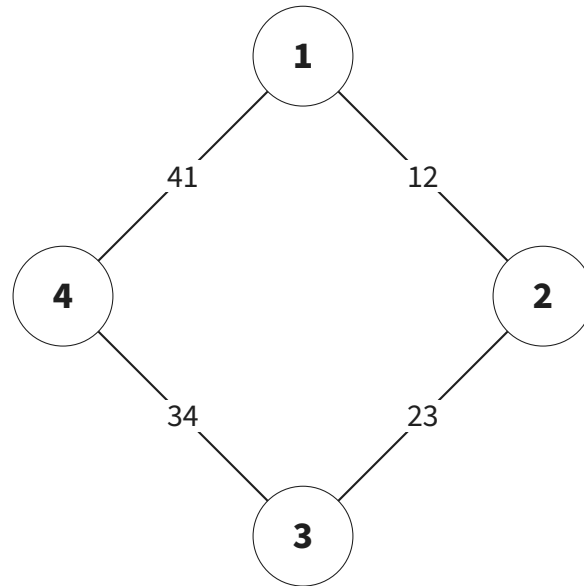
Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

Revised September 27, 2026. Unpiloted revisions; record actual use after teaching.

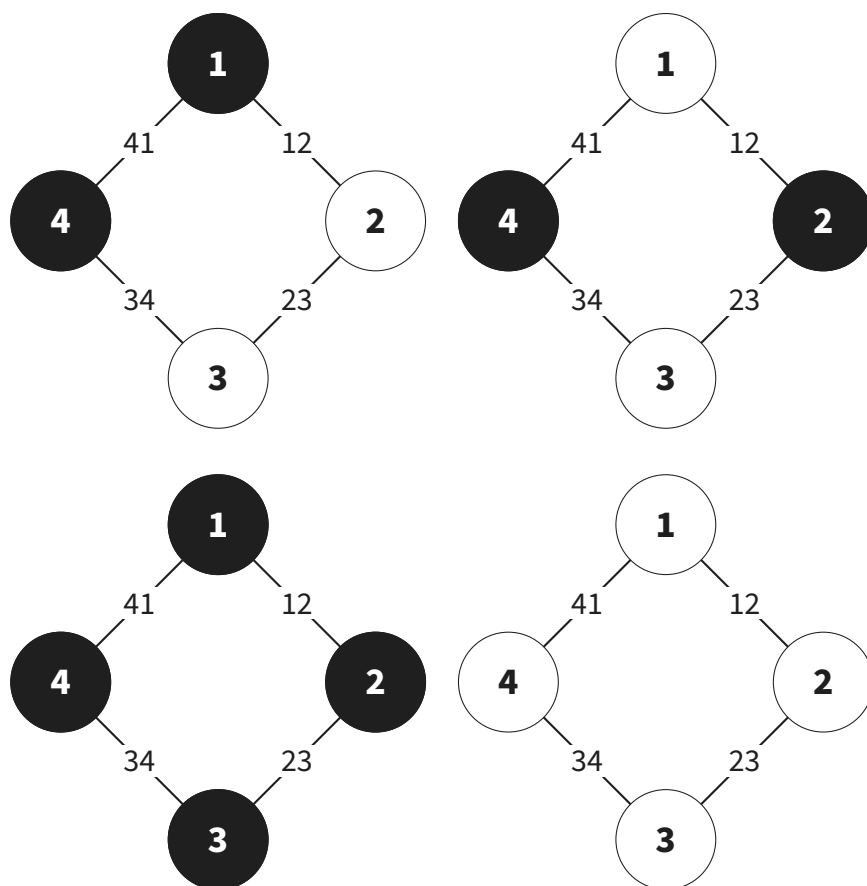
Problem 1: Put a two-sided counter on each lamp. White is OFF; dark is ON. Start all OFF. Choose a line and flip both counters at its ends. Flip that same pair again. Try another line and its undo.



Problem 2: Start all OFF each time. Make only lamps 1 and 2 ON. Then make only lamps 1 and 3 ON. Point to the lines you used.

Draw one attempt you want to remember.

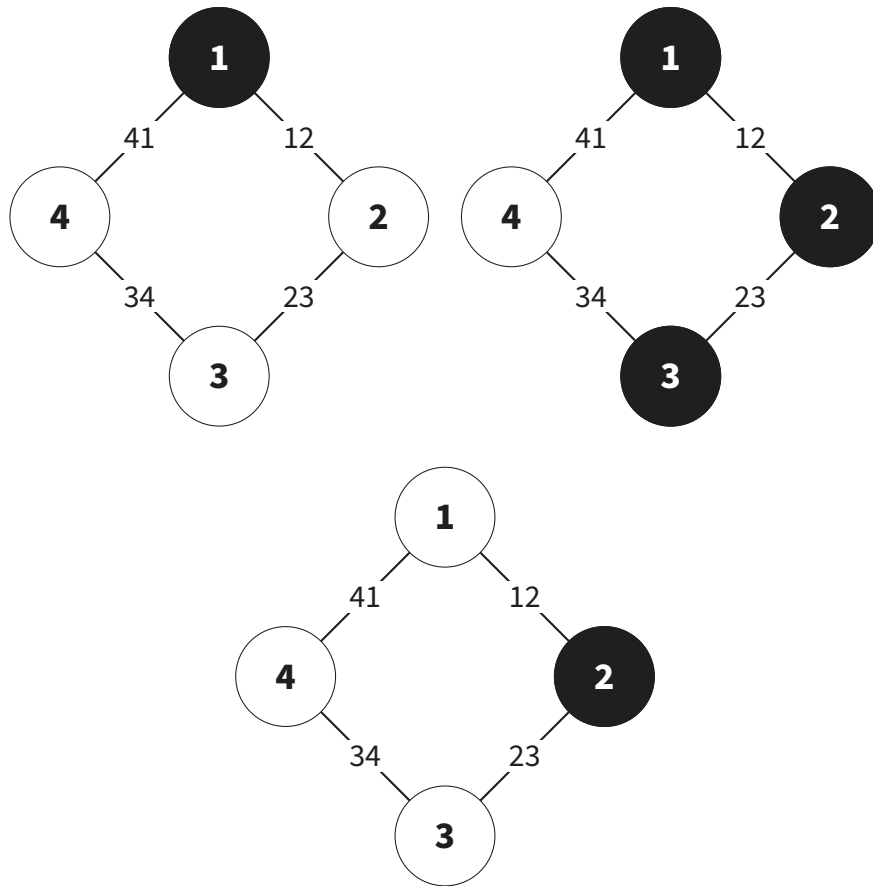
Problem 3: Start all OFF for each picture. Flip the two ends of a line each move. Make the dark lamps ON and the white lamps OFF. Mark a picture when you make it.



Problem 4: Choose a picture you made. Make it again. Undo your moves to return to all OFF. Show someone your moves in both directions.

Draw or mark the picture you chose.

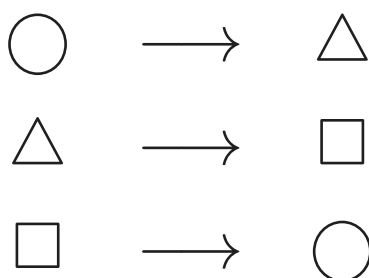
Problem 5: Start all OFF. Try to make each picture using the two-end rule. Circle any picture you make. Leave the others uncircled for now.



Problem 6: Choose one uncircled picture. Change the rule: you may also flip just lamp 1. Try your picture again. Show which move uses the new rule.

Draw the picture you made with the new rule.

Problem 1: One turn changes each symbol using this key. Keep its place in the message. Point to an input and follow its arrow. Try each symbol.



Problem 2: Send each message through the key. Draw or place its new symbols in the space beside it.

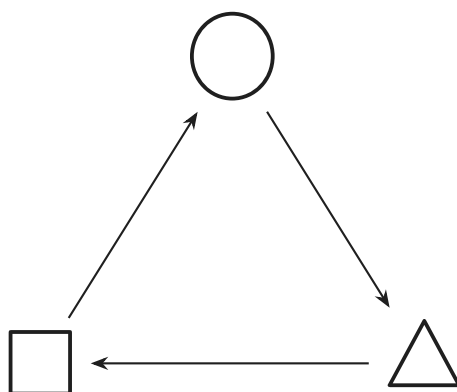
Message	New message
○ △ ○	
□ □ △	
△ ○ □	

Problem 3: A friend received triangle, circle, triangle. Follow the key backward to find the original message. Check by sending it forward again.

Problem 4: Use the same key. Start with a circle card. After every turn draw what it becomes. Keep going until the circle first comes back. Repeat from triangle and from square.

Start	Turn 1	Turn 2	Turn 3
○			
△			
□			

Problem 5: These arrows show the same key as a loop. Move a counter one arrow each turn. Try all three starting shapes. Mark each arrival at its starting shape.



Problem 6: Start with circle, triangle, square in a row. Change all three each turn. Stop when the whole row is back. Draw the row just before it comes back.

Problem 7: Change the key: circle and triangle swap; square stays square. Try all three starting shapes. Draw their results for two turns.

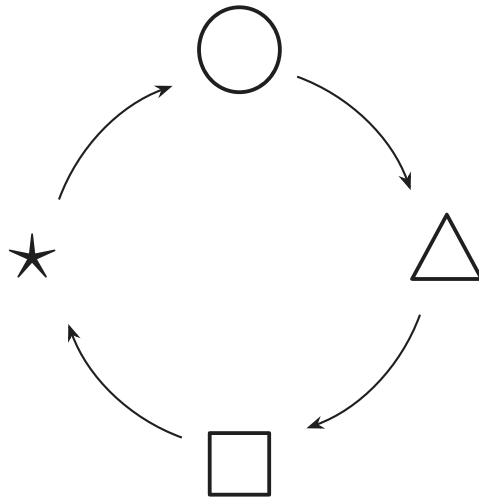


Start	Turn 1	Turn 2
		
		
		

Problem 8: Start with all three shapes in a row. Repeat this new key until the whole row returns. Compare with the old key. Record the number of turns for each.

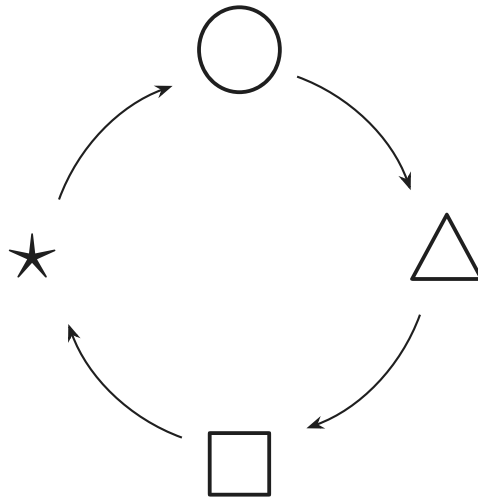
Problem 9: Try another key: circle becomes square, and triangle also becomes square. Secretly choose circle or triangle and show only its output. Can your partner tell which input you chose? Swap roles and try again. Draw the two possible inputs.

Problem 1: Put a counter on circle. One hop follows an arrow to the next shape. Move one hop at a time until you return to circle. Draw each place you land, ending with circle.



Problem 2: Start at triangle. Take one hop each turn until you return. Draw the landings. Compare with your route from circle.

Problem 3: Start at circle. Now take two hops every turn. Draw only the shape where each turn ends. Stop when circle returns. Repeat from triangle.



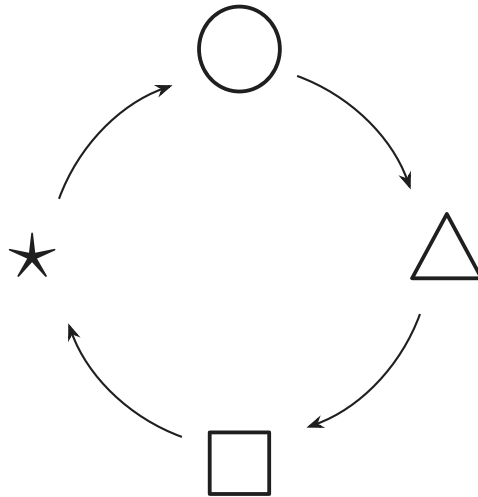
From circle:

From triangle:

Problem 4: Try to land on triangle by taking two hops each turn from circle. Then change your starting shape to square and try again. Circle the shapes you can reach from these two starts.



Problem 5: Start at circle. Take three hops every turn. Draw each landing until circle returns. Compare the shapes visited by one-hop, two-hop, and three-hop turns.



Problem 6: A secret turn moves every counter the same number of clockwise hops, fewer than four. Circle becomes square. Use the ring to find where triangle, square, and star go. Draw their outputs.

Problem 7: Try one turn that undoes that secret turn. Test it from each of the four starting shapes. Show someone each out-and-back trip.

Problem 1: Use towers of heights 1, 2, and 3. Put them in the order 2, 1, 3 from left to right. Look from the left, then the right. A taller tower hides a shorter one behind it. Write 2, 1, 3 from left to right in both rows below. Circle the towers you see from each end.

Looking from the left

leftright

Looking from the right

leftright

Problem 2: Rearrange the towers into 1, 3, 2. Look from both ends again. Draw or write the tower heights. Circle the towers you see.

Looking from the left

leftright

Looking from the right

leftright

Problem 3: Rearrange the same three towers for each view below. Draw or write each row from left to right.

Towers seen from the left	My tower order
3	
1	
2	

Problem 4: Build different rows with heights 1, 2, and 3. Keep the left end in the same place. Record each new order and count the towers seen from each end. Do not record the same order twice.

Row	Tower order, left to right	Seen left	Seen right
1			
2			
3			
4			
5			
6			

Problem 5: Compare your recorded rows. Circle the rows that show two towers from the left and two from the right. Build each circled row again and check both views.

A row I checked:

Problem 6: Try to make a row of heights 1, 2, and 3 that shows only one tower from each end. Record two different attempts. For each attempt, count from both ends.

Tower order	Seen left	Seen right

Problem 7: Make a tiny city using two towers of height 1 and two of height 2. Put one tower in each square. Every row and every column must contain both heights. Record two different cities. Circle the city whose top row shows two towers from the left.

Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Keep the secret RB visible. Build each test below. Point to matching positions, then record the total.

Test	Score against secret RB
RR	
RB	
BR	
BB	

Problem 2: Hide a two-place secret. Your partner makes tests and you give scores. Keep testing until your partner can name the secret. Record tests below or on scrap paper. Reveal the secret and check the scores. Swap jobs and play again.

First secret: _____

Next secret: _____

Test	Score

Test	Score

Problem 3: Build the four possible secrets RR, RB, BR, BB. Test RR against each one and record its score. Circle every secret that could remain if the answer were 1.

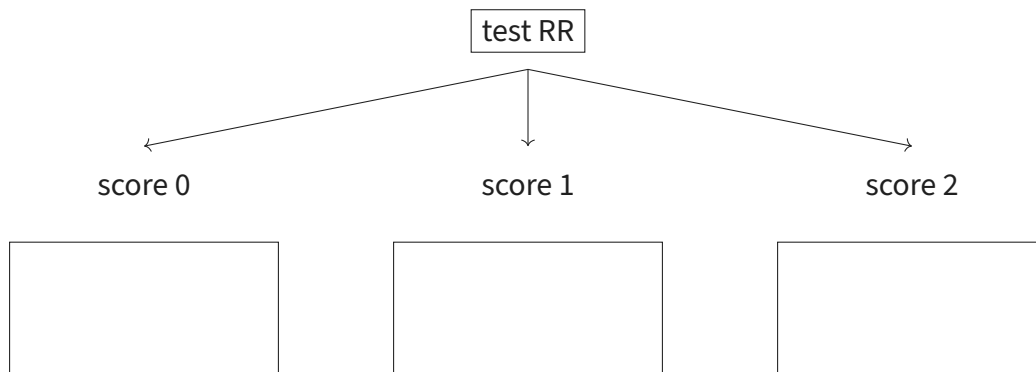
Possible secret	Score for test RR
RR	
RB	
BR	
BB	

Problem 4: Keep just the two circled secrets. Try RR again, then try RB. Record the score each test would receive for each remaining secret. Circle a test that tells the secrets apart.

Test	Score if secret is RB	Score if secret is BR
RR		
RB		

A different test that also tells them apart:

Problem 5: Begin with test RR. Under each answer, write or build every secret that could remain. Use your checks from Problem 3.



Problem 6: If a branch has two possible secrets, choose a test and record its score for each secret. If a branch has one secret, name it without another test.

My next test and what each answer means:

Problem 7: Ask your partner to hide each of RR, RB, BR, and BB in an order you do not know. Follow your branches. Record how many tests you needed before you could name each secret.

Secret after reveal	RR	RB	BR	BB
Number of tests				

Problem 1: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Play from 4 counters twice, swapping who starts. Draw or write the number taken at each turn. Circle the last move.

Game 1 moves:

--

Game 2 moves:

--

Problem 2: Play from 5 counters twice, swapping who starts. Record the moves and circle the last move. Use the same 1-or-2 rule.

Game 1 moves:

--

Game 2 moves:

--

Problem 3: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Start with 3 counters. Let the other player start. Try each of their first moves below. Record how many you take next.

Starting pile	Other player takes	I take next	Counters left
3	1		
3	2		

Problem 4: Start with 6 counters. Make two groups of 3. Let the other player start. Try each first move below and choose your reply so that exactly 3 counters remain. Then finish each game.

Other player takes	I take next	Counters left
1		
2		

Draw the two groups before the game:

Problem 5: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Try 7, 8, and 9 counters. For each pile, choose whether to go first or second. If you go first, record your first move. Play twice from each pile and check your choice.

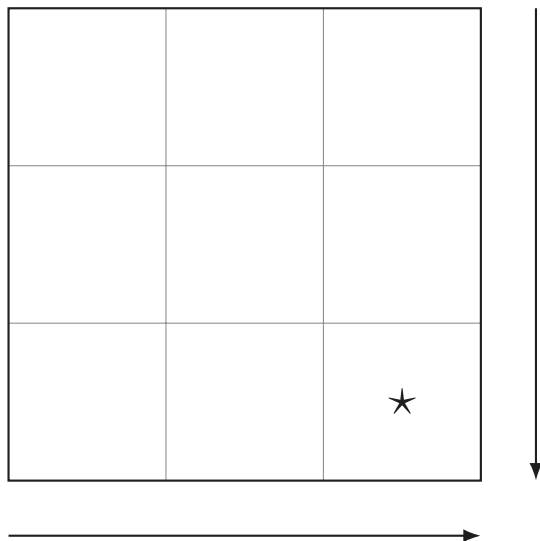
Pile	I choose first or second	My first move if I start
7		
8		
9		

Problem 6: Build 12 counters as four groups of 3. Let the other player start. After each of their moves, take enough to remove 3 counters between your two turns. Record the numbers left after each of your turns. Play until the counters are gone.

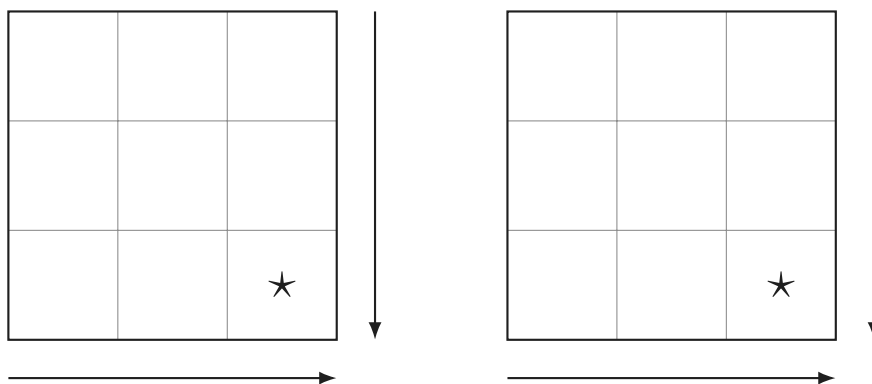
Draw my four groups:

Counters left after my turns:

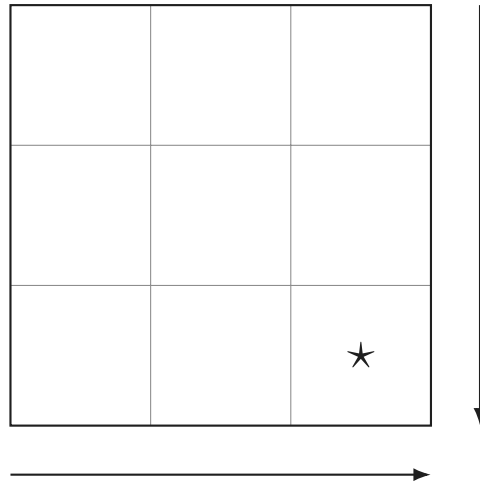
Problem 1: Share one token. Take turns moving it right or down. Move one or more squares; do not move diagonally. Whoever reaches the star wins. Play from the square just above the star, then from the middle square. Switch who starts and play both again.



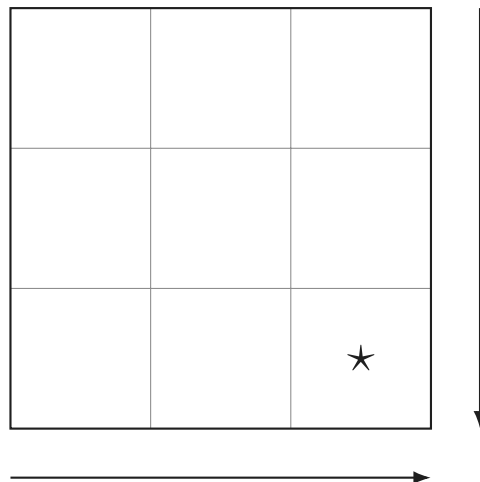
Problem 2: Put the token in the middle. Draw each possible first move on a small board. Then draw the next player's move to the star.



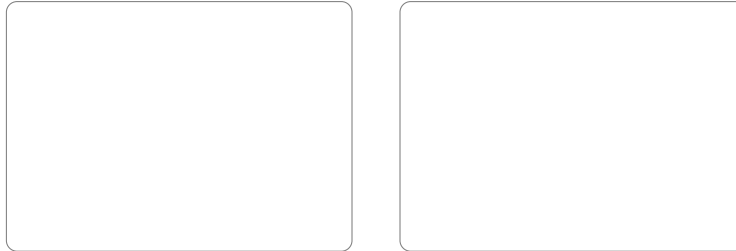
Problem 3: Start in the top-left square. Play twice, switching who starts. Try a long first move in one game and a short first move in the other. Draw a first move and its reply below.



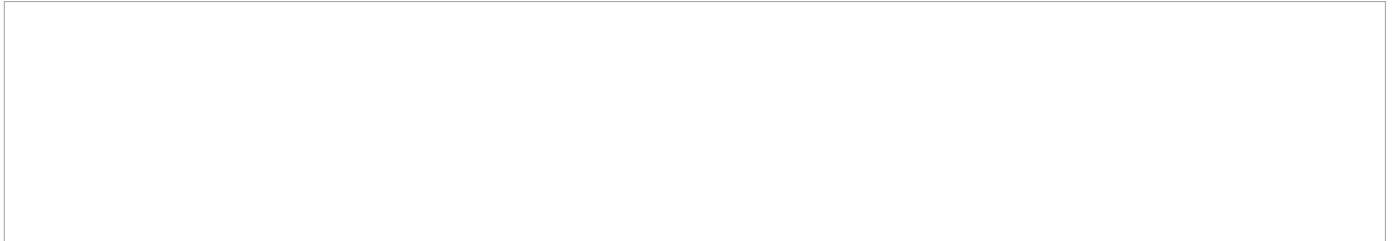
Problem 4: Now start in the left square of the middle row. Play twice, switching who starts. Choose a first move that leaves the same distance right and down to the star. Draw that move.



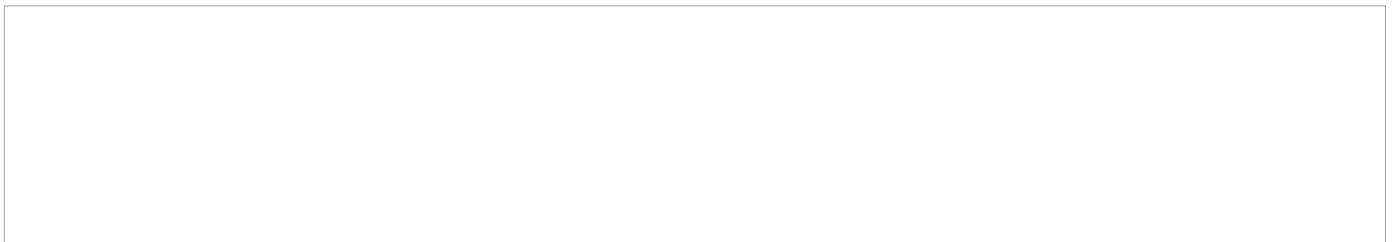
Problem 5: Build two piles of 2 counters in these spaces. Take turns removing one or more counters from just one pile. Whoever takes the last counter wins. Play twice, switching who starts.

Two empty rectangular boxes, each intended for a pile of 2 counters.

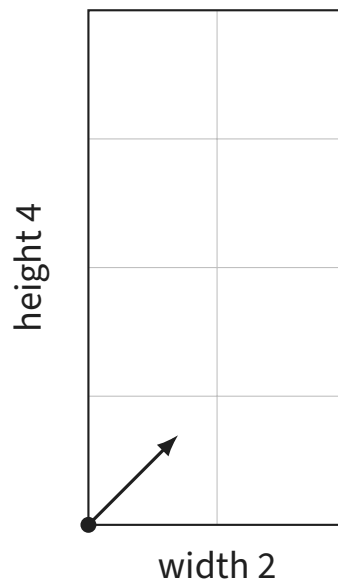
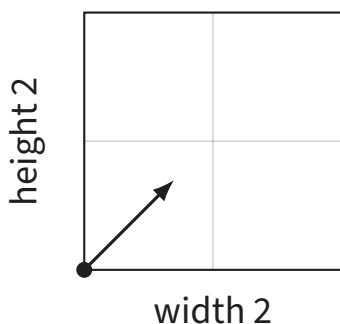
Problem 6: Build two piles of 3. After your partner takes counters, take the same number from the other pile. Keep playing that way. Draw the piles after one pair of turns.

A large empty rectangular box for drawing the piles after one pair of turns.

Problem 7: Build piles of 3 and 1. Choose a first move that leaves equal piles. Keep playing. Then choose two new unequal piles and ask a partner to make them equal in one move. Draw your new piles.

A large empty rectangular box for drawing the new piles.

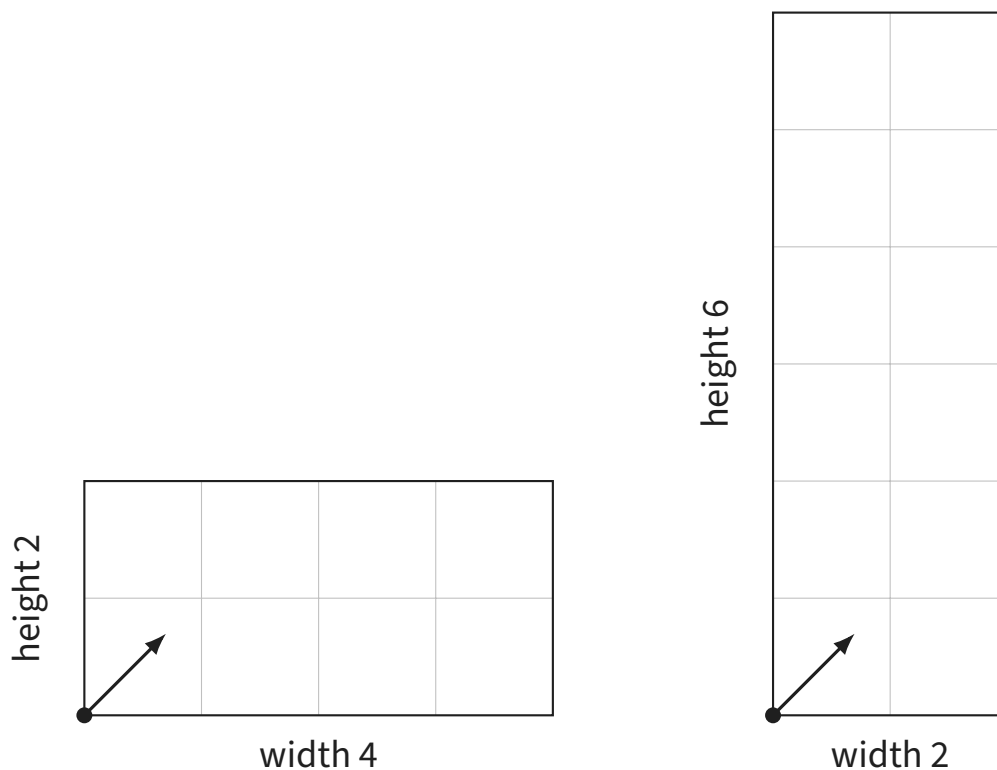
Problem 1: Move a counter from the black dot diagonally up and right. At a side wall turn left or right; at the top or bottom turn down or up. Keep following the diagonals. Stop at the first corner. Try both tables. Draw the paths and circle the ending corners.



Problem 2: Move along the tall table again. At its right wall, show two possible next directions with your finger. Choose the one that stays inside and keeps going up. Draw that turn below.

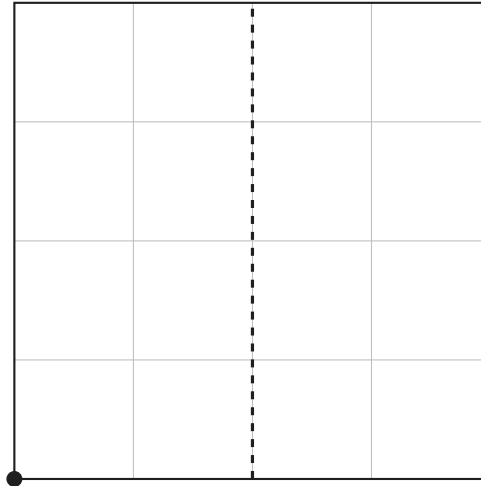
A large empty rectangular box for drawing the path and turning on the tall table.

Problem 3: Predict where each path will end. Put a small dot at your prediction. Trace the path from the black dot, bouncing at walls and stopping at the first corner. Circle the corner you reach.

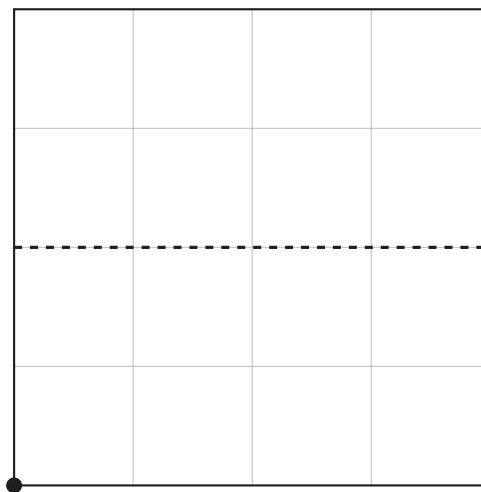


Problem 4: Choose two tables on these pages that finish at different corners. Move a counter along each. Draw the final piece of each path below.

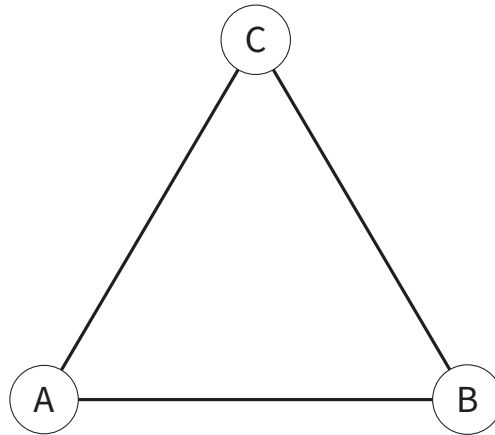
Problem 5: The dashed line below is the wall between two mirror copies of the tall table. Draw one straight diagonal from the black dot to the far corner. Fold on the dashed wall. Trace the two pieces of your line as a bouncing path.



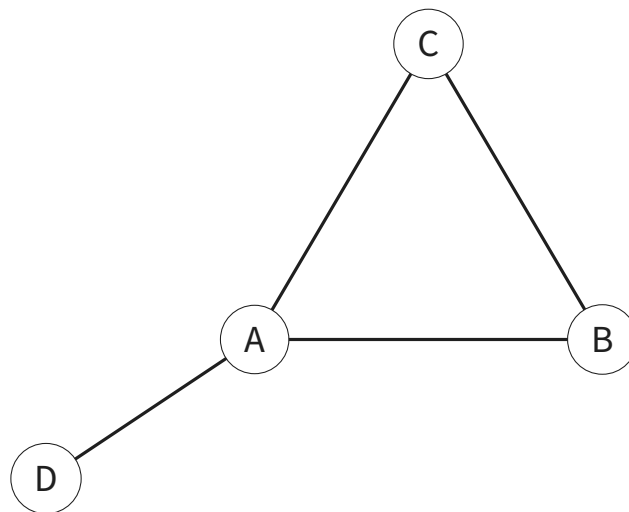
Problem 6: These are mirror copies of the wide table. Draw the same straight diagonal, then fold on this dashed wall. Trace the bouncing path and point to its ending corner. Compare it with Problem 5.



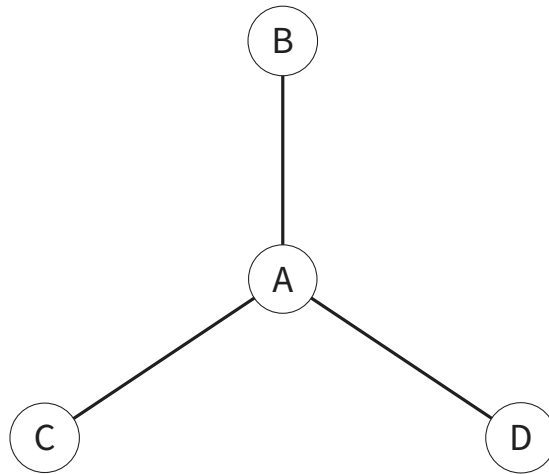
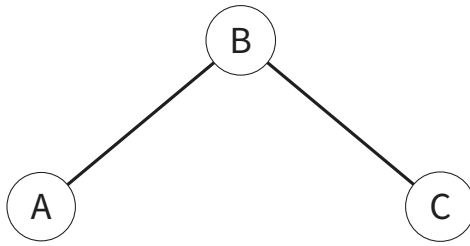
Problem 1: Dots are islands; lines are bridges. Move a counter along every bridge once without jumping. You may visit an island again. Mark a bridge after you use it. Try this town from A, then from B. Draw arrows for one complete walk.



Problem 2: This town has one extra bridge. Try a walk from D, then try from B. Mark used bridges as you go. Circle a place where you can start a walk using every bridge once.

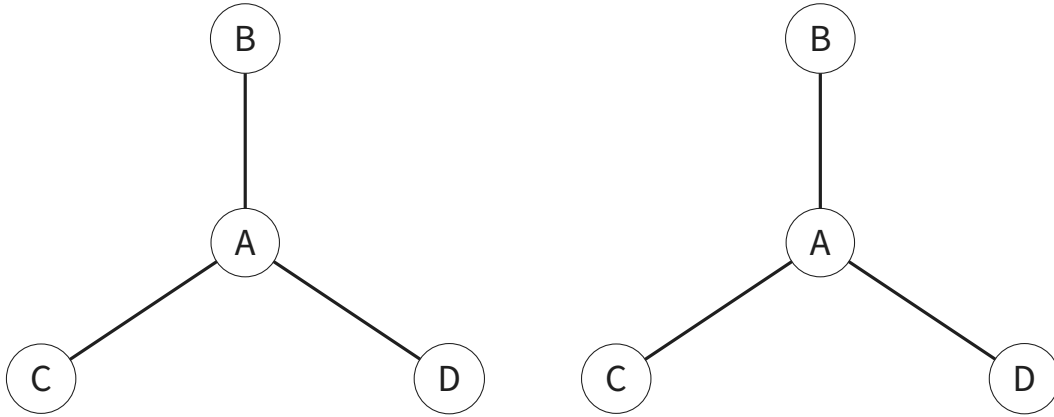


Problem 3: Try a walk using every bridge once in each town. Mark used bridges. Circle where a complete walk starts and ends. If you get stuck, reset and try a different start.



Problem 4: On the three-branch town, try starting at A and then at B. Draw your two attempted walks below. Mark the bridges left unused.

Problem 5: Add one bridge between two outside islands in the left town. Try a walk using every bridge once. Add a different bridge to the right town and try again. Number the bridges in walking order or draw arrows.



Problem 6: Choose one repaired town. Circle its starting and ending islands. Put a small mark on the island with only one bridge. Move the counter along the complete route again.

Problem 7: Draw a small town of your own with three or four islands. Add bridges, then try to use every bridge exactly once. Change a bridge if needed. Show a partner the walk.